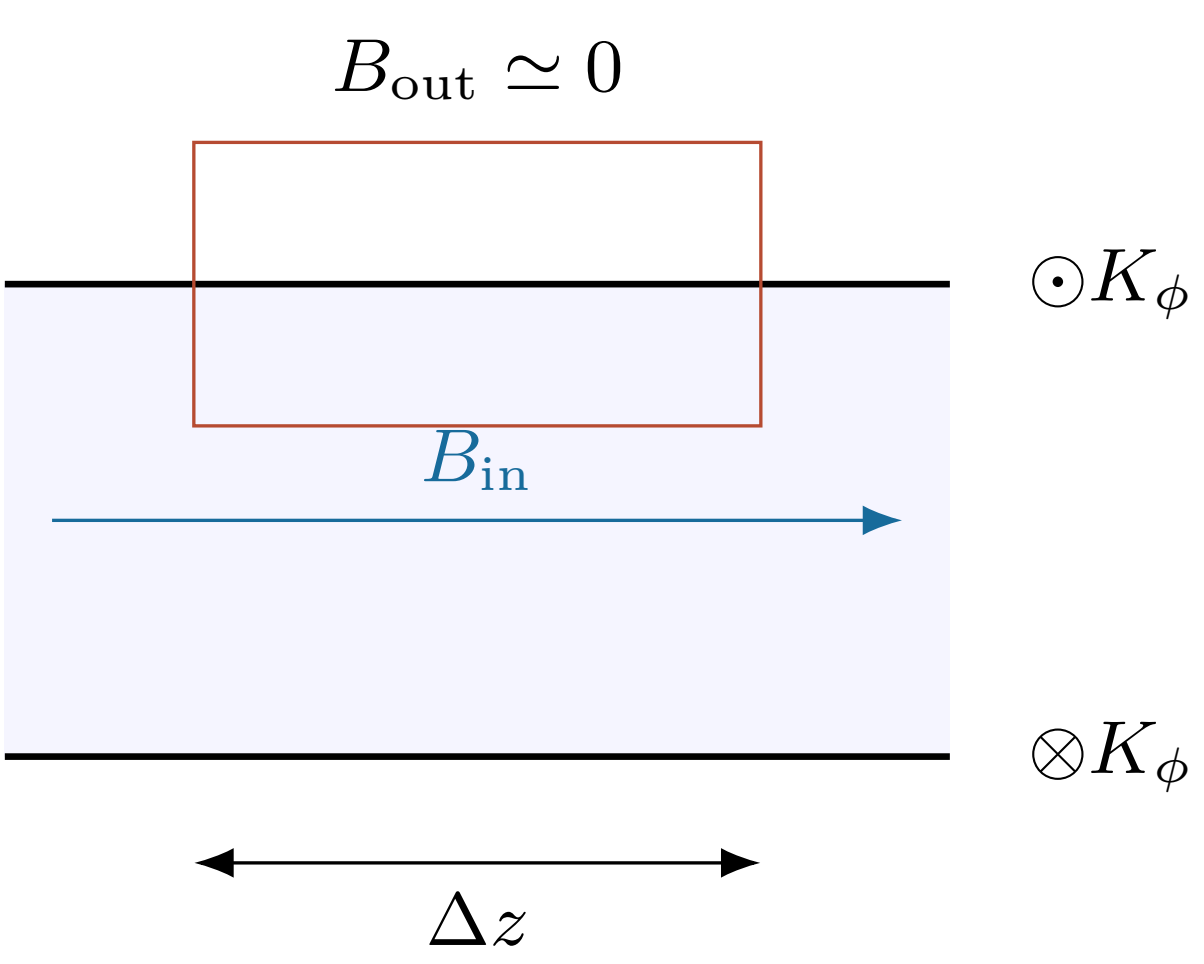

$$B_{\text{in}} - B_{\text{out}} = \mu_0 K_\phi$$